JUNIOR SCHOOL LEARNING HUB

GRADE 9 INTEGRATED SCIENCE
END-OF-TERM 2 EXAMINATION

Name: _____	Adm. No.: _____	Date: _____	Class: _____
Time: 2 hours 30 minutes	Total: 100 marks	Term: 2	Learning Area: Integrated Science

INSTRUCTIONS TO LEARNERS

- Answer all questions in Section A and Section B.
- Write clear, concise answers and show calculations where required.
- Use labelled diagrams where they help to explain your answers.
- Observe all safety precautions when describing or carrying out practical activities.

SECTION A: SHORT-ANSWER QUESTIONS (40 MARKS)

1. State two differences between series and parallel electrical circuits. (2 marks)

2. Name the instrument used to measure electric current in a circuit. (2 marks)

3. State the SI unit of electrical resistance. (2 marks)

4. A current of 2 A flows through a resistor of 6 Ω. Calculate the potential difference across it. (2 marks)

5. Name the process by which heat is transferred through a vacuum. (2 marks)

6. State two ways of reducing heat loss from a building. (2 marks)

7. Define the term density. (2 marks)

8. A body has a mass of 200 g and a volume of 50 cm^3. Calculate its density. (2 marks)

9. State two characteristics of an image formed by a plane mirror. (2 marks)

10. Name the part of the human eye that controls the amount of light entering it. (2 marks)

11. State two functions of the skeleton in humans. (2 marks)

12. Name the blood component responsible for transporting oxygen. (2 marks)

13. State the function of the kidney in the human body. (2 marks)

14. Name the organelle where aerobic respiration mainly takes place. (2 marks)

15. State two ways in which microorganisms are useful to humans. (2 marks)

16. Name the gas released during photosynthesis. (2 marks)

17. State two conditions necessary for seed germination. (2 marks)

18. What is meant by the term ecosystem? (2 marks)

19. Give two examples of renewable sources of energy. (2 marks)

20. State two laboratory safety precautions when heating substances. (2 marks)

SECTION B: STRUCTURED QUESTIONS (60 MARKS)

21. Electricity and electrical safety (10 marks)

a) Draw a simple circuit containing a cell, a switch, a bulb and connecting wires. (3 marks)

b) State the function of a fuse in an electrical circuit. (2 marks)

c) A resistor has a resistance of 4 Ω and is connected to a 12 V supply. Calculate the current flowing through it. (3 marks)

d) Give two safety precautions when using electrical appliances. (2 marks)

22. Heat transfer and temperature (10 marks)

a) Name the three main methods of heat transfer. (3 marks)

b) Explain how heat from the Sun reaches the Earth. (2 marks)

c) Explain why a metal spoon placed in hot tea becomes hot at the handle. (3 marks)

d) State two features of a good cooking pot that improve heat transfer to food. (2 marks)

23. Human body systems (10 marks)

a) Name the four main components of blood. (4 marks)

b) State one function of each of the following: red blood cells and white blood cells. (2 marks)

c) Describe the pathway of air from the nose to the lungs. (2 marks)

d) State two ways of maintaining a healthy respiratory system. (2 marks)

24. Respiration, photosynthesis and plant growth (10 marks)

a) Write the word equation for photosynthesis. (3 marks)

b) State two differences between photosynthesis and aerobic respiration. (2 marks)

c) State three conditions required for successful seed germination. (3 marks)

d) Explain why a plant kept in darkness for a long time becomes weak and pale. (2 marks)

25. Matter, measurement and density (10 marks)

A solid metal block has a mass of 540 g. Its dimensions are 10 cm × 3 cm × 2 cm.

a) Calculate the volume of the block. (2 marks)

b) Calculate its density. (3 marks)

c) Explain why the block sinks when placed in water. (2 marks)

d) State three precautions when measuring the volume of an irregular solid by water displacement. (3 marks)

26. Environment, energy and microorganisms (10 marks)

a) Distinguish between renewable and non-renewable energy sources. (2 marks)

b) Give two examples of each type of energy source. (2 marks)

c) Explain two roles of decomposers in an ecosystem. (2 marks)

d) State two useful applications of microorganisms. (2 marks)

e) State two measures that can reduce water pollution in the community. (2 marks)

MARKING SCHEME

Grade 9 Integrated Science – End-of-Term 2 Examination

Total: 100 marks | Section A: 40 marks | Section B: 60 marks

SECTION A (40 MARKS)

1. Series: one path/current same throughout; parallel: branches/more than one path and components operate independently. Any two clear contrasts. (2)

2. Ammeter. (2)

3. Ohm (Ω). (2)

4. V = IR = 2 × 6 = 12 V. (2)

5. Radiation. (2)

6. Any two: roof/wall insulation, double glazing, draught-proofing, thick curtains, reflective roofing. (2)

7. Mass per unit volume of a substance. (2)

8. Density = 200/50 = 4 g/cm³. (2)

9. Any two: virtual, upright, same size as object, laterally inverted, same distance behind mirror as object is in front. (2)

10. Iris. (2)

11. Support, protection, movement, blood cell production. Any two. (2)

12. Red blood cells/haemoglobin in red blood cells. (2)

13. Filters blood, removes wastes and regulates water/salt balance by producing urine. (2)

14. Mitochondrion/mitochondria. (2)

15. Any two: fermentation, decomposition, food production, medicine production, nitrogen fixation. (2)

16. Oxygen. (2)

17. Any two: water, oxygen, suitable temperature. (2)

18. A community of organisms interacting with one another and with the non-living environment. (2)

19. Any two: solar, wind, hydroelectric, geothermal, biomass, tidal. (2)

20. Any two: use holder/tongs; point test tube away; wear goggles; keep flammables away; do not leave flame unattended. (2)

SECTION B (60 MARKS)

21. Electricity and electrical safety (10 marks)

Award 1 mark each for a correct cell, switch and bulb connected in a complete series circuit with wires. (3)

A fuse melts and breaks the circuit when excessive current flows, helping to prevent damage/fire. (2)

V = IR; I = V/R = 12/4 = 3 A. (3)

Any two: do not handle with wet hands; avoid damaged cords; switch off before repairs; do not overload sockets; use proper earthing. (2)

22. Heat transfer and temperature (10 marks)

Conduction, convection and radiation. (3)

By radiation; it does not require a material medium. (2)

Heat travels through the metal by conduction, from the hot end towards the cooler handle. (3)

A good conducting base/material; broad, flat base for greater contact area; close-fitting lid to reduce heat loss. Any two. (2)

23. Human body systems (10 marks)

Plasma, red blood cells, white blood cells and platelets. (4)

Red blood cells transport oxygen; white blood cells defend the body against pathogens. (2)

Nose/nasal cavity → trachea → bronchi → bronchioles → alveoli/lungs. (2)

Avoid smoking; avoid dusty/polluted air; exercise; maintain good hygiene; seek treatment for infections. Any two. (2)

24. Respiration, photosynthesis and plant growth (10 marks)

Carbon dioxide + water —(light and chlorophyll)→ glucose + oxygen. (3)

Photosynthesis stores energy while respiration releases energy; photosynthesis uses carbon dioxide and releases oxygen while respiration uses oxygen and releases carbon dioxide; photosynthesis occurs in chloroplasts while aerobic respiration mainly occurs in mitochondria. Any two contrasts. (2)

Water, oxygen and suitable temperature. (3)

It cannot photosynthesise adequately; little chlorophyll/food is produced, causing pale, weak growth. (2)

25. Matter, measurement and density (10 marks)

Volume = 10 × 3 × 2 = 60 cm^3. (2)

Density = mass/volume = 540/60 = 9 g/cm^3. (3)

Its density (9 g/cm^3) is greater than the density of water (about 1 g/cm^3). (2)

Read the meniscus at eye level; lower the object gently; ensure it is fully submerged; remove air bubbles; record initial and final readings accurately. Any three. (3)

26. Environment, energy and microorganisms (10 marks)

Renewable sources are naturally replenished; non-renewable sources are finite or replaced very slowly. (2)

Renewable: solar, wind, hydro, geothermal, biomass. Non-renewable: coal, petroleum, natural gas, uranium. Any two correctly classified. (2)

Break down dead organisms/waste and recycle nutrients into the environment. (2)

Food fermentation, medicine/antibiotic production, decomposition, sewage treatment, nitrogen fixation. Any two. (2)

Treat sewage/industrial waste; dispose of refuse properly; avoid dumping chemicals; protect water sources; use appropriate sanitation. Any two. (2)

ADMINISTRATION NOTE

This is an original teacher-prepared assessment resource for practice and school-based assessment. Before administration, the Integrated Science teacher should verify that the assessed concepts match the actual Grade 9 Term 2 teaching sequence and the school scheme of work. Accept scientifically correct equivalent responses and award method marks where indicated.

Prepared by: Junior School Learning Hub